

PRACTICE TASKS

TASK 1 — CHECK YOUR IP ADDRESS

Objective

Find:

- IPv4 Address
- Default Gateway
- DNS Server

Steps

Step 1

Press: Windows + R

Step 2 Type: cmd

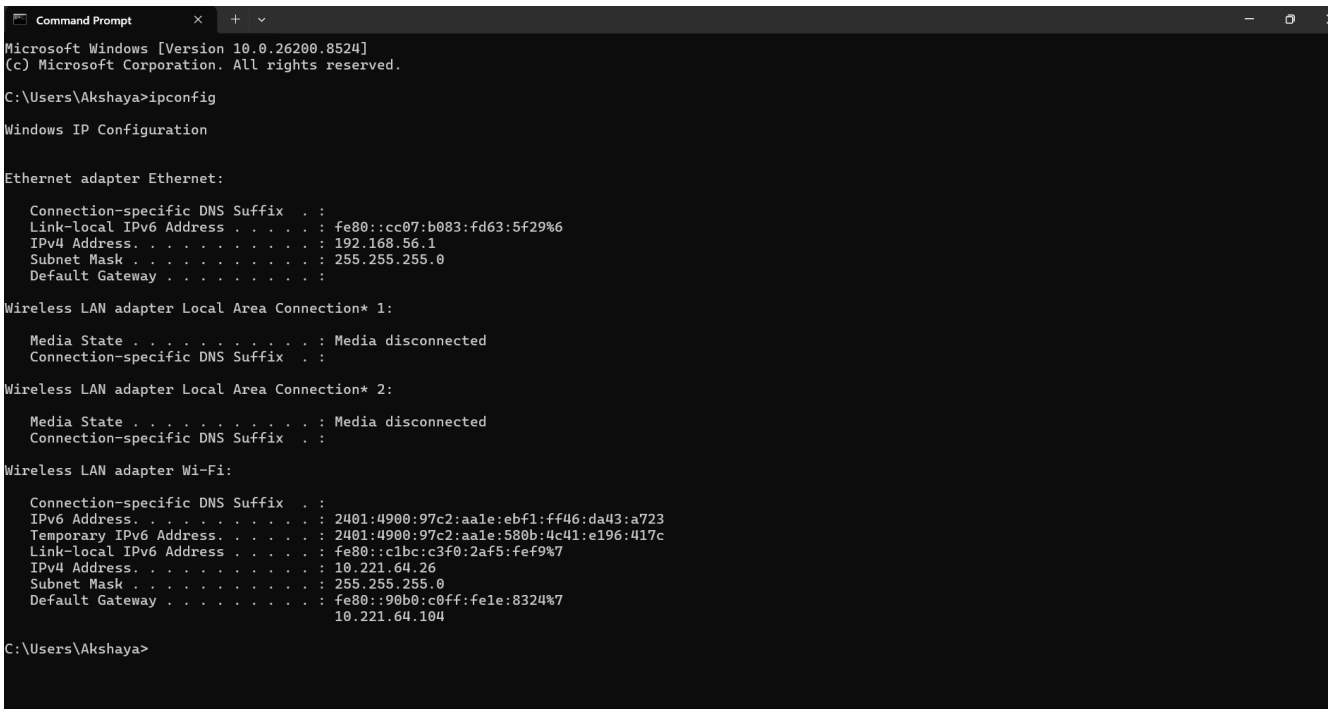
Press Enter.

Step 3 Command Prompt opens.

Type: ipconfig

Press Enter.

Screenshot



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-Local IPv6 Address . . . . . : fe80::cc07:b083:fd63:5f29%6
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:97c2:aa1e:ebf1:ff46:da43:a723
    Temporary IPv6 Address. . . . . : 2401:4900:97c2:aa1e:580b:4c41:e196:417c
    Link-Local IPv6 Address . . . . . : fe80::c1bc:c3f0:2af5:fef9%7
    IPv4 Address. . . . . : 10.221.64.26
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::90b0:c0ff:fe1e:8324%7
                                10.221.64.104

C:\Users\Akshaya>
```

NOTES

- **IPv4 Address** Identifies your device in network.

- **Default Gateway** Router address used to access internet.
- **DNS Server** Converts domain names into IP addresses.

TASK 2 — TEST INTERNET CONNECTIVITY

Objective

Check whether internet is working.

Steps

In Command Prompt type:

ping google.com

Press Enter.

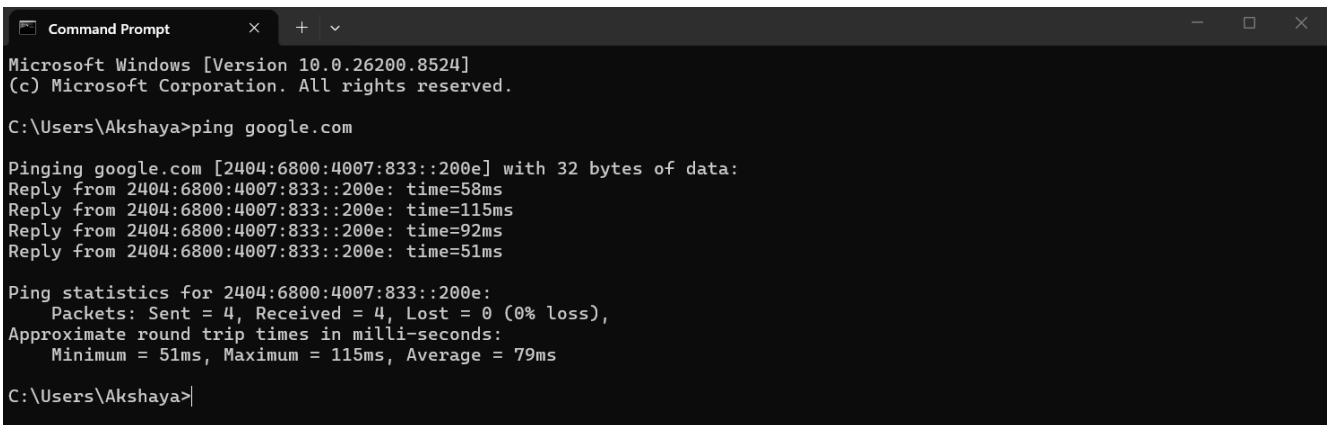
Learn

Reply Packets: Shows connection success.

Time: Measures response speed.

Packet Loss: Shows network problems.

Screenshot



```
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ping google.com

Pinging google.com [2404:6800:4007:833::200e] with 32 bytes of data:
Reply from 2404:6800:4007:833::200e: time=58ms
Reply from 2404:6800:4007:833::200e: time=115ms
Reply from 2404:6800:4007:833::200e: time=92ms
Reply from 2404:6800:4007:833::200e: time=51ms

Ping statistics for 2404:6800:4007:833::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 51ms, Maximum = 115ms, Average = 79ms

C:\Users\Akshaya>
```

Ping command checks:

- Connectivity
- Network availability
- Response time

TASK 3 — DNS LOOKUP

Objective

Understand how DNS converts names into IPs.

Steps

Type:

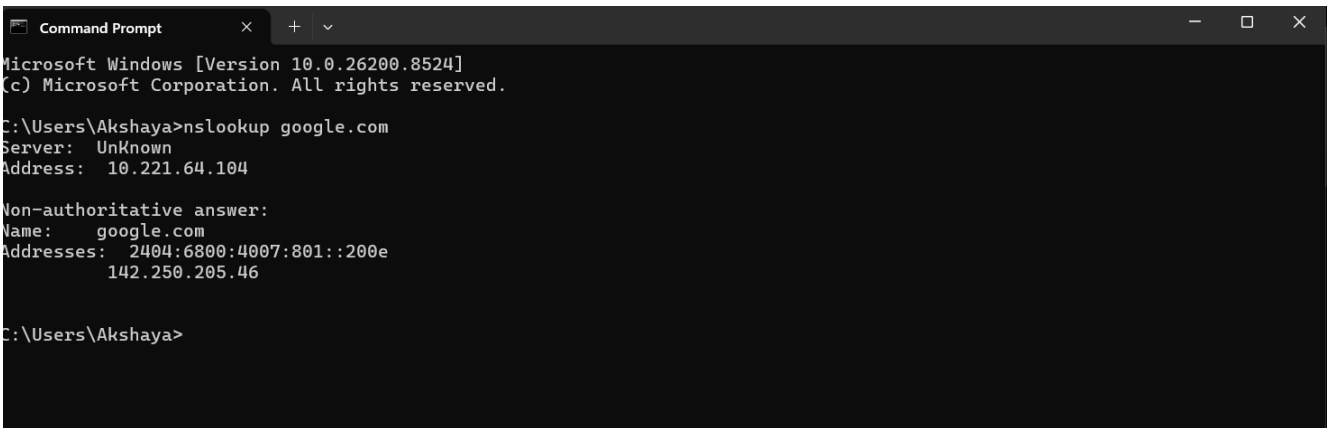
nslookup google.com

Press Enter.

Learn

DNS converts: google.com into : 142.x.x.x

Screenshot



```
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>nslookup google.com
Server:      UnKnown
Address:     10.221.64.104

Non-authoritative answer:
Name:        google.com
Addresses:   2404:6800:4007:801::200e
             142.250.205.46

C:\Users\Akshaya>
```

DNS helps users access websites using names instead of remembering IP addresses.

TASK 4 — IDENTIFY PRIVATE IP

Objective

Check whether your IP is private.

Steps

Look at your IPv4 from:

ipconfig

Compare With Private Ranges

192.168.x.x
10.x.x.x
172.16.x.x – 172.31.x.x

Example

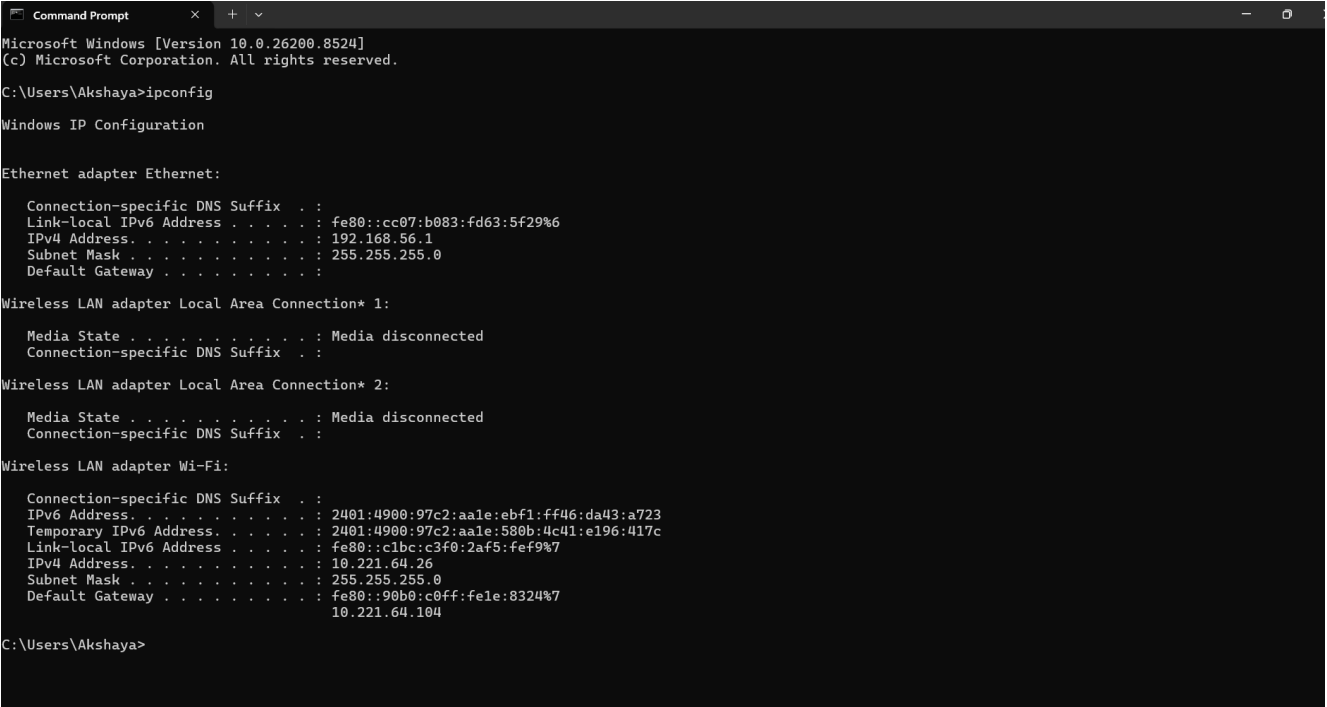
If your IP is:

192.168.1.5

Then it is a:

Private IP

Screenshot



```
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::cc07:b083:fd63:5f29%6
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
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Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:97c2:aale:ebf1:ff46:da43:a723
    Temporary IPv6 Address. . . . . : 2401:4900:97c2:aale:580b:4c41:e196:417c
    Link-local IPv6 Address . . . . . : fe80::c1bc:c3f0:2af5:fef9%7
    IPv4 Address. . . . . : 10.221.64.26
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::90b0:c0ff:fe1e:8324%7
                                10.221.64.104

C:\Users\Akshaya>
```

Private IP addresses are used inside local networks and cannot directly access the internet.

TASK 5 — WRITE 5 VALID IPv4 ADDRESSES

Objective

Understand IPv4 structure.

Steps

Write any 5 valid IPv4 addresses.

Example:

192.168.1.1
10.0.0.5
172.16.1.20
8.8.8.8
192.168.10.25

IMPORTANT NOTES

IPv4 contains:

- 4 octets
- Values from 0–255

TASK 6 — RESEARCH PUBLIC IP

Objective

Understand Public IP vs Private IP.

Steps

Open browser.

Search:

What is my IP

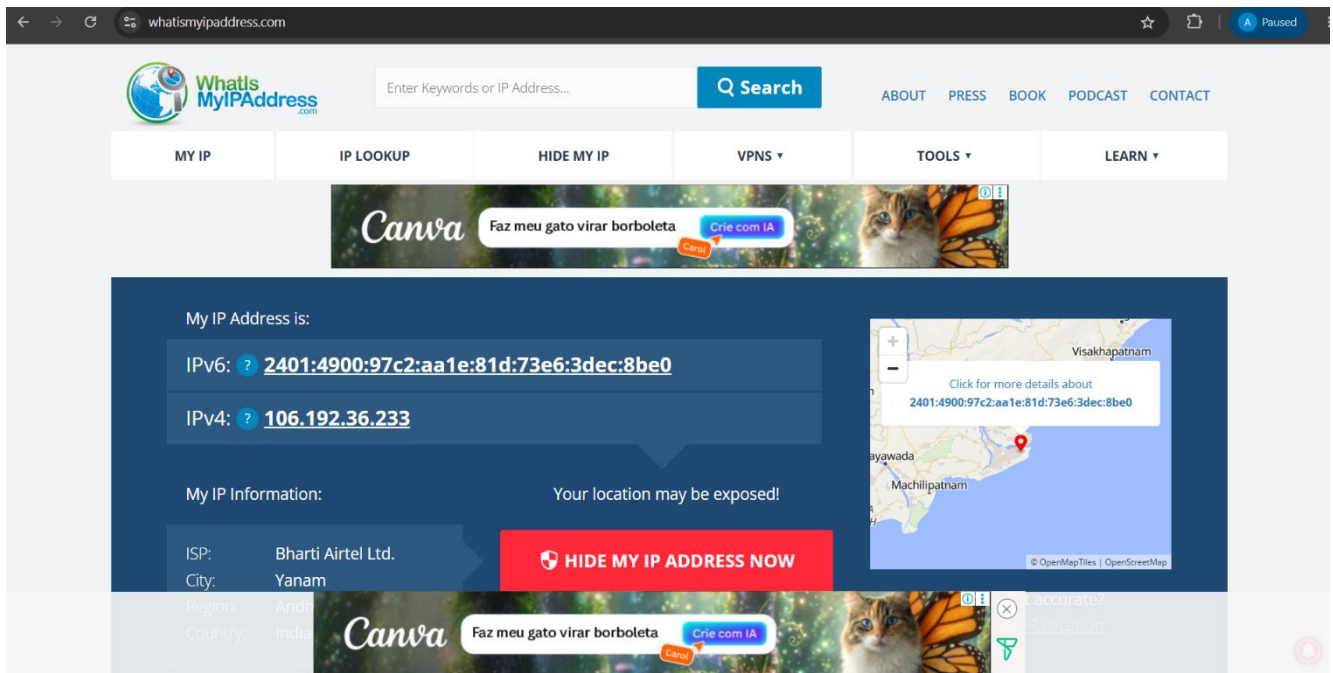
Google shows your public IP.

Learn

Public IP: Used on internet.

Private IP: Used inside local network.

Screenshot



The screenshot shows the website 'What's My IP Address' (whatismyipaddress.com) in a browser. The page displays the user's public IP address as 2401:4900:97c2:aa1e:81d:73e6:3dec:8be0 (IPv6) and 106.192.36.233 (IPv4). It also shows the user's location as Visakhapatnam, India, with a map. The page includes a search bar, navigation links (ABOUT, PRESS, BOOK, PODCAST, CONTACT), and a 'HIDE MY IP ADDRESS NOW' button. There are also advertisements for Canva and a 'Crie com IA' button.

Public IP is assigned by ISP and visible on internet.

TASK 7 — COMMON TERMS

Objective

Learn basic networking terminology.

Write Meanings

IP: Unique address for devices.

DNS: Converts names into IP addresses.

Gateway: Connects network to internet.

Router: Routes traffic between networks.

Network: Group of connected devices.

TASK 8 — IAM CONNECTION TASK

Objective

Understand networking importance in IAM.

Write:

“How IP Addresses Help IAM Security”

Minimum 5 points.

Example Points

1. Track user login locations
2. Detect suspicious logins
3. Apply conditional access
4. Restrict unauthorized access
5. Monitor authentication activities